Kirtan Patel

Chicago, IL | +1 (773) 701-0710 | official.kirtan13@gmail.com | linkedin.com/in/kirtanpatel2003 | github.com/kirtanpatel2003

Skills

Languages & Scripting: Python, Go, Bash, SQL, JavaScript
Systems & Networking: Linux, Windows, MacOS; TCP/IP, DNS, HTTP/S, common ports & protocols; Cisco switch troubleshooting (classroom networking)
IT Support & Infrastructure: TeamDynamix (TDX), tier-2 escalation, classroom networking, AV & lecture-capture (Echo360), hardware diagnostics & repair, printer fleet management
Security Tools: Nmap, Burp Suite, Metasploit, sqlmap, Hashcat, BloodHound, Impacket, Wireshark (Hack The Box)
Development: ReactJS, Next.js, Node.js, Flask, PostgreSQL, MongoDB, Git, Docker

Open Source

llm-thorn | *Python, FastAPI, httpx, Pydantic, SQLite, PyPI* *Author & Maintainer*

- Authored and published (PyPI: `llm-thorn`) a runtime semantic security layer that sits as a reverse proxy in front of any LLM, screening every request and response through layered detectors – heuristic, semantic intent, multi-turn context risk, and output analysis – governed by a YAML policy-as-code engine with five enforcement actions (allow, warn, block, redact, terminate).
- Built a tamper-evident, hash-chained audit log with a CLI integrity-verification command for compliance-sensitive use; stood up full CI/CD from scratch (CodeQL, secret scanning, typed async internals via FastAPI and Pydantic v2, Docker image), staying backend-agnostic across cloud and local providers (OpenAI, Anthropic, Ollama).

Red_Co-Author | *Python, Ollama, Streamlit, scikit-learn, Laminar* *Author*

- Engineered a fully-local Python red-team and defensive monitor stack for the Co-Authoring Jailbreak (CoJP) LLM vulnerability, automating a two-armed attack pipeline (direct prompt vs CoJP-framed prompt) across multiple local target models orchestrated through Ollama and scoring both responses with a separate judge model on the HarDBench 1-to-5 harmfulness rubric.
- Trained a defensive monitor on the resulting attack dataset using local nomic-embed-text embeddings plus a scikit-learn logistic regression classifier; evaluated for generalization on held-out AdvBench (520 prompts) and HarDBench (400 prompts) splits and benchmarked head-to-head against a zero-shot llama3 baseline. Surfaced the full pipeline through a Streamlit dashboard with ASR-by-target heatmaps and ROC curves.

Curio (urban-toolkit/curio) | *Python, Flask, Socket.IO, React, TypeScript* *Credited Contributor*

- Designed and built a real-time, multi-device collaboration layer for Curio, a dataflow framework for urban visual analytics, that lets multiple users edit the same workflow together over a shared network; presented the working feature to the project owner and core collaborators.
- Implemented the sync model on Socket.IO: live presence with per-user color coding, per-node soft locks with "Editing: name" badges, accept/decline code-change proposals with first-accept-wins conflict resolution, and execution-output propagation across collaborators' downstream nodes.
- Maintainers adopted the prototype into a `collaboration-revised` branch; the event vocabulary and proposal/lock model from the prototype ship in the released feature, with credit by name in the project's official documentation at urban-toolkit/curio/docs/COLLABORATION.md.

Projects

Hunter | *Python, Textual, Ollama, httpx, SQLite, pytest* *In Progress*

- Building a local-first, evidence-gated AI assistant for authorized penetration testing that runs a staged recon-to-report workflow fully offline, treating every model claim as a hypothesis that must be cross-checked against raw tool output before it can enter a report.
- Engineered a verification layer that extracts hard tokens (ports, versions, CVEs) from model output and cross-checks them against observed evidence, discarding and logging hallucinations; enforced a hard scope fence beneath a bounded autonomy loop (capped iterations, wall-clock, and actions) so no probe can touch an out-of-scope target.

Mini Tor Anonymity Network | *Python, TCP Sockets, Curve25519, AES-256-GCM, HKDF, SHA-256*

- Implemented Tor's three-hop onion routing protocol from scratch as six independent processes (directory, guard/middle/exit relays, client, destination) communicating over real TCP, with telescoping EXTEND cells so the client only ever connects to the guard while middle and exit are reached through the growing circuit.
- Built the entire cryptographic stack in pure Python with no third-party crypto libraries: SHA-256 (FIPS 180-4), HMAC, HKDF (RFC 5869), X25519 Diffie-Hellman via Montgomery ladder (RFC 7748), AES-256 (FIPS 197), and AES-256-GCM (NIST SP 800-38D), each verified against published NIST and IETF test vectors.

Ethos OS Access Control Checker | *Go, Ethos OS, Mandatory Access Control*

- Extended the initial permissions configuration of Ethos, a research operating system built at UIC around label- and group-based mandatory access control, defining a user group and protection label so that only group members were authorized to write a protected file.
- Built a Go checker that resolves a file's label at runtime, loads the permission structure from the kernel's `/auth` hierarchy, evaluates group membership, and compares the resulting policy decision against actual kernel behavior.
- Testing across authorized and unauthorized users showed the kernel granting writes the policy denied, isolating a gap where label/group enforcement is wired for IPC but not for ordinary file I/O.

DiffFence | *Python, OSV.dev, Ollama*

- Built a local-first Python CLI that acts as a diff-aware security gate, scanning code changes for security vulnerabilities and production risks before a push or merge.
- Implemented baseline-on-first-run logic so subsequent scans surface only new findings, eliminating repeated noise and keeping developer focus on incremental risk introduced by each change.

Work Experience

Technology Solutions, Office of the Vice Chancellor for Innovation – UIC Chicago, IL
IT Helpdesk Support Manager *Dec 2024 – Present*

- Lead the IT helpdesk support team, owning tier-2 escalation in TeamDynamix (TDX) across classroom networking, AV systems, and campus-wide technology while coaching specialists through complex tickets to resolution.
- Assess and provision faculty, staff, and student devices and access, applying least-privilege and safe-configuration judgment to hardware, software, and account provisioning across campus.
- Co-author and maintain the team's troubleshooting documentation, translating recurring technical issues into clear, repeatable resolution steps that keep support consistent and shorten new-hire ramp-up.
- Serve on the hiring team for support specialists, interviewing and evaluating candidates for technical skill and the ability to support non-technical users, then onboarding new hires onto the team.

Technology Solutions, Office of the Vice Chancellor for Innovation – UIC Chicago, IL
Classroom Technology Support Specialist *Aug 2022 – Dec 2024*

- Kept instruction running across East Campus classrooms and lecture halls, supporting AV systems, projectors, audio, and Echo360 lecture capture and resolving live issues quickly to protect class time.
- Deployed and configured computer labs for courses and campus events, handling software installs and access provisioning.
- Trained and mentored incoming specialists to independent troubleshooting through shadowing, hands-on practice, and feedback on real tickets.

Education

University of Illinois Chicago (UIC) Chicago, IL
Master of Science in Computer Science *December 2026*

- Relevant Coursework: Network Security, Cryptography, Computer System Security, Wearable Technology, Data Visualization & Analytics

University of Illinois Chicago (UIC) Chicago, IL
Bachelor of Science in Computer Science – Software Engineering Concentration; Minor in Mathematics *May 2025*

- Dean's List: College of Engineering – Fall 2023 & Fall 2024
- Relevant Coursework: Building Secure Computer Systems, Artificial Intelligence, Database Systems, Software Engineering 1 & 2, UI/UX, Software Design, Statistical Methods, Data Structures